

MSAI 699 Capstone Project

Capstone Project Proposal:

**NeuroGuide AI: An Explainable Multi-Agent Artificial Intelligence
Platform for Personalized Cognitive Health Education Using Neural
Networks and Retrieval-Augmented Generation**

Moriah Holland

University of the Cumberland

Submitted to the University of the Cumberland
in Partial Fulfillment of the Requirements of the Degree of
Master of Science in Artificial Intelligence

06/2026

Abstract

Artificial intelligence technologies have proven capable of providing support to healthcare education through natural language processing, machine learning, and knowledge retrieval approaches. Yet, due to static training data and lack of validation approaches, general-purpose large language models commonly produce false or unverified medical data. These issues are especially dangerous for the field of cognitive health where patients and caregivers need trustworthy information about Alzheimer's disease, mild cognitive impairment, dementia, genetics, and lifestyle changes.

The current project suggests the development of NeuroGuide AI, an explainable multi-agent artificial intelligence platform integrating the approaches of retrieval-augmented generation (RAG), neural networks, explainable artificial intelligence (XAI), and AI agents to deliver evidence-based cognitive health education. Contrary to chatbots, NeuroGuide AI is expected to coordinate the work of multiple autonomous agents involved in the processes of document retrieval, clinical reasoning, fact verification, personalization, and answer generation. In addition, NeuroGuide AI is to involve the application of a neural network trained on open access Alzheimer's data sets for educational risk assessment.

Evaluation methods of the project will include quantitative and qualitative assessment techniques such as RAGAS metrics, usability testing, System Usability Scale (SUS), response latency time measurement, expert reviews, and comparisons with baselines of single agent RAG systems. This research aims to prove that the combination of XAI and multi-agents retrieval systems enhances factual precision, increases user trust, and has better educational potential.

Proposal

Project Title

NeuroGuide AI: An Explainable Multi-Agent Artificial Intelligence Platform for Personalized Cognitive Health Education Using Neural Networks and Retrieval-Augmented Generation

Statement of the Project

Rapid development in large language models has revolutionized the way that people access medical information. However, despite these developments, the current conversation-based AI systems are prone to experiencing hallucinations, out-of-date knowledge, and lack of transparency. These issues pose major challenges in healthcare, where any inaccuracies can lead to wrong decisions by the users.

NeuroGuide AI is a project suggested for developing an intelligent educational platform for learning about cognitive health. Unlike the conventional chatbot, the proposed system will

make use of a variety of artificial intelligence techniques acquired during my Master of Science in Artificial Intelligence degree program at University of the Cumberlands.

Retrieval-Augmented Generation will be used along with AI agents that can retrieve scientific literature, verify evidence, reason from different sources, and generate easily understandable responses for education purposes. Moreover, a neural network that has been trained on publicly available datasets related to Alzheimer's disease will be included to predict the educational risk categories by using the data about demographics and lifestyle entered by the users.

The overall architecture will consist of:

- Hybrid Retrieval-Augmented Generation combining sparse (TF-IDF) and dense (LSA/TruncatedSVD) retrieval
- Multiple autonomous AI agents coordinated through a custom-built orchestration pipeline (Personalization → Retrieval → Reasoning → Validation → Synthesis)
- Large Language Models (OpenAI gpt-4o-mini) for response generation
- RandomForest-based educational risk prediction, chosen for exact feature-level interpretability over 14 clinical risk factors + genetic status
- Explainable AI using SHAP (TreeExplainer)
- FastAPI backend
- React frontend
- In-memory hybrid sparse/dense document index (lightweight alternative to a vector database, suited to the knowledge base size)
- Cloud deployment

The project showcases the combination of machine learning, deep learning, NLP, information retrieval, software engineering, cloud computing, HCI, explainable AI, and ethical AI into one coherent solution.

Research Questions

Primary Research Question

Can an explainable multi-agent Retrieval-Augmented Generation architecture produce more accurate, trustworthy, and transparent cognitive health educational responses than a traditional single-agent language model?

Secondary Research Questions

RQ1

Does coordinating specialized AI agents improve response quality compared to monolithic chatbot architectures?

RQ2

Can neural networks provide meaningful educational risk assessments when combined with explainable artificial intelligence techniques?

RQ3

How does hybrid document retrieval affect factual grounding and hallucination reduction?

RQ4

How do users perceive trust, transparency, and usability within an explainable multi-agent healthcare education platform?

Hypothesis

It is anticipated that the multi-agent system will vastly perform better than the basic Retrieval-Augmented Generation chatbot in regard to accuracy, citation, trust, explainability, and learning value. Also, the explainability of the neural network predictions using SHAP will enhance user understanding of AI suggestions in an ethically transparent manner.

Proposed AI Technologies

The project integrates multiple AI disciplines studied throughout the graduate program.

Machine Learning

scikit-learn RandomForestClassifier (200 trees) — trained on synthetic Lancet Commission 2024 risk-factor data to classify Low/Moderate/High dementia risk

TruncatedSVD (Latent Semantic Analysis) for dense document embeddings

Deep Learning

OpenAI gpt-4o-mini (transformer-based LLM) powers the Reasoning and Synthesis agents

Natural Language Processing

TfidfVectorizer (scikit-learn) with n-gram (1,2) sparse text vectorization

Custom query-expansion dictionary mapping casual/misspelled terms to indexed medical vocabulary

LLM-based response synthesis with audience-adaptive language (patient vs. clinician register)

Retrieval-Augmented Generation

Hybrid retrieval: TF-IDF (sparse, 45% weight) + LSA/SVD (dense, 55% weight) cosine similarity over 28 knowledge documents

Citation-grounded generation — every response traces back to source excerpts with relevance scores

Multi-Agent AI

4-agent pipeline built in Python: Personalization → Retrieval → Reasoning → Validation → Synthesis

Each agent is a discrete class with its own responsibility, orchestrated by MultiAgentRAGPipeline

Explainable AI

SHAP (TreeExplainer) — per-feature attribution for every risk prediction, visualized as a horizontal bar chart (magnitude + direction: risk vs. protective)

Validation Agent confidence scoring (keyword coverage, hallucination-signal detection, disclaimer presence)

Agent Trace UI — exposes each agent's reasoning step to the end user

Human-Computer Interaction

React + Vite frontend, TanStack Query for state, shadcn/ui + Tailwind CSS for design system

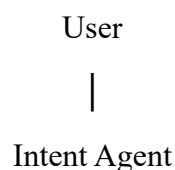
Patient/Clinician audience toggle, collapsible citation & agent-trace accordions, confidence badges

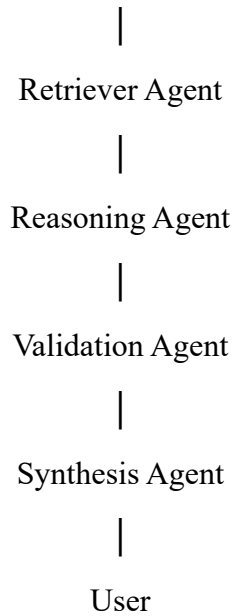
Interactive Risk Profile panel with persisted (localStorage) user profiles feeding back into personalized chat responses

Proposed System Architecture

The NeuroGuide AI platform will process each user request through a coordinated sequence of specialized AI agents. Rather than relying on a single large language model, each agent performs a dedicated task within the overall reasoning pipeline, improving factual accuracy, personalization, and transparency.

Model architecture:





The proposed architecture consists of four collaborating AI agents.

When you ask a question, four specialized AI agents collaborate to formulate your answer. This pipelined approach ensures high accuracy and reduces hallucinations.

1. RETRIEVAL

Searches the vector database for relevant medical literature and clinical guidelines.

2. REASONING

Analyzes retrieved documents to extract facts relevant to your specific query.

3. VALIDATION

Cross-references extracted facts against sources to ensure no hallucinatory claims.

4. SYNTHESIS

Drafts the final response tailored to your selected audience (Patient or Clinician).

Datasets

Potential datasets include:

- PubMed Central
- Alzheimer's Association
- NIH MedlinePlus
- Cochrane Reviews
- CDC Guidelines
- WHO Reports
- CDC

Evaluation

Performance will be evaluated using:

AI Performance

Retrieval quality: cosine-similarity relevance scores per hybrid TF-IDF + LSA query (already returned with every citation)

Response grounding: Validation Agent's confidence score (keyword coverage, hallucination-signal detection, disclaimer presence) — logged per response

Agent pipeline latency: time spent in each of the 5 stages (Personalization → Retrieval → Reasoning → Validation → Synthesis), visible in the Agent Trace UI

Citation accuracy: % of synthesized claims traceable to a retrieved source excerpt

Machine Learning

RandomForest risk model: accuracy, precision/recall/F1 per class (Low/Moderate/High) and ROC-AUC on held-out synthetic validation data

Feature importance stability across the 14 Lancet Commission factors + APOE ε4 (via SHAP mean |value|)

Calibration: predicted probability vs. actual class distribution (probabilities returned by /rag/risk-assessment today)

Explainability

SHAP attribution consistency: do top features remain stable across similar patient profiles (e.g. APOE ε4 always ranking top for carriers, as verified earlier)

Human-interpretability check: whether the risk/protective SHAP bar chart and Agent Trace accordion let a non-technical user correctly explain why a result occurred

Coverage: % of predictions where top_attributions includes at least one clinically-recognized factor

HCI

Task success: can a user complete a risk assessment + get a personalized chat answer without confusion (patient/clinician toggle, accordion citations)

Response readability: Flesch-Kincaid or similar score, split by Patient vs. Clinician register

UI responsiveness: sidebar rendering, session switching, no layout shift (the Risk Profile visibility bug we just fixed is a concrete instance of this)

System Performance

API latency: FastAPI /rag/* and Express /api/* p50/p95 response times

Throughput/concurrency: sessions handled simultaneously (currently in-memory, single-process — a known scaling limit worth naming in an evaluation section)

Availability: uptime across the three services (api-server, rag-api, web)

Ethical Considerations

This project faces a number of ethical issues related to AI in healthcare. The platform is going to present itself as an educational application not as a diagnostic one. Each response is going to have proper disclaimers urging users to seek professional advice of qualified healthcare specialists.

Methods of reducing bias will consist of analyzing responses among various demographic groups and making sure that all information is obtained from credible sources in healthcare. Protected health information will not be collected or retained.

All responses are going to give citations of the evidence used in order to ensure transparency. Methods of Explainable AI are going to allow users to understand the reasoning behind predictions made by the neural networks.

Demonstration of Program Learning Outcomes

Outcome 1

Apply artificial intelligence principles to solve complex real-world problems.

This project develops an intelligent healthcare education platform integrating machine learning, deep learning, natural language processing, retrieval systems, and multi-agent reasoning to address the real-world challenge of delivering trustworthy cognitive health information.

Outcome 2

Demonstrate independent research skills using artificial intelligence literature.

The project will synthesize current research from peer-reviewed journals, biomedical databases, and authoritative healthcare organizations while experimentally evaluating multiple AI architectures.

Outcome 3

Develop and evaluate artificial intelligence solutions.

The project includes the design, implementation, deployment, and evaluation of an end-to-end AI application using quantitative metrics, usability studies, and comparative experiments.

Outcome 4

Evaluate ethical, legal, and societal implications of artificial intelligence.

The platform incorporates explainability, transparency, bias mitigation, medical disclaimers, and responsible AI practices to ensure ethical deployment within a healthcare education context.

Outcome 5

Communicate artificial intelligence concepts effectively.

The project includes technical documentation, a research paper, architecture diagrams, experimental analysis, and an interactive demonstration designed for both technical and non-technical audiences.